

Sheaf-Indexed Code Search: Using the Semantic Site for Retrieval

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Abstract

We introduce SHEAFSEARCH, a code search system using JU GEO’s semantic site as a structured index. Unlike syntactic (grep) or neural retrieval, SHEAFSEARCH represents both code-base and query as sheaves over semantic sites, measuring similarity via *covering compatibility* and *stalk matching*. The `semantic_closure` API expands queries, `formal_core_site` builds the index, and `discovery_pipeline` orchestrates retrieval. Retrieved code satisfies matching type constraints, control flow, and trust annotations. Experiments on 18 programs show site construction in 0.05 ms, closure in 0.00 ms, and full pipeline in 0.00 ms with 100% success rate.

1 Introduction

Code search is fundamental to software development. Existing tools are either syntactically fast but semantically blind (grep, ripgrep) or neurally aware but opaque and unverifiable (CodeBERT, Copilot). Neither leverages the *verified semantic structure* of code—type signatures, invariants, pre/postconditions, and trust levels form a rich profile that tokens and embeddings only approximate.

Consider a developer searching for “a function that sorts a list while preserving its length, with solver-level verification.” Syntactic search returns every function mentioning “sort”; neural search may rank results by embedding similarity but cannot guarantee that returned code actually satisfies the length-preservation postcondition. What is needed is a search system that operates on the *verified semantic structure* of programs—one that can match not merely on names and tokens, but on the mathematical content of specifications, the topology of covering relations, and the trust level of evidence.

The semantic site as index. In the JUGEO framework, every program P is associated with a semantic site $\mathcal{S}_P = (\text{Ob}(\mathcal{S}_P), \text{Cov}(\mathcal{S}_P))$. Objects are *coordinates*—typed program components carrying specifications—and coverings express how local judgments assemble into global properties. The judgment sheaf $\mathcal{G}$ over $\mathcal{S}_P$ assigns each coordinate its full 8-tuple $(c, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$, encoding type, propositions, evidence, obligations, obstructions, trust, and provenance. Across our benchmark of 18 programs, sites average 6.5 coordinates and 14.2 morphisms, providing a rich relational index that goes far beyond flat feature vectors.

Our approach. SHEAFSEARCH treats the semantic site as a first-class search index. Each corpus program has a precomputed site $\mathcal{S}_P$; a query is expressed as a site $\mathcal{S}_Q$. Retrieval reduces to finding site morphisms from $\mathcal{S}_Q$ into corpus sites.

The key insight is that code search can be understood as a problem in *sheaf theory*: a query defines a small site with constraints on its sections, and a valid search result corresponds to a site morphism $f: \mathcal{S}_Q \rightarrow \mathcal{I}$ such that pulling back the judgment sheaf yields sections compatible with the query constraints. This formulation gives us three advantages: (1) compositionality, since queries decompose along covers; (2) soundness, since morphism existence implies logical entailment; and (3) completeness, since the index contains all intra-program morphisms.

Running example. Throughout this paper we use a concrete query: find all functions of type `List[int] → List[int]` that satisfy `is_sorted(result)` and `len(result) = len(input)`, with trust at least SOLVER. In our corpus this query should return sorting functions (merge sort, insertion sort, etc.) while excluding functions that sort but truncate, or that preserve length but do not sort.

Contributions. We present (1) a formal *sheaf-indexed search* model grounded in Grothendieck topology (§4), (2) categorical foundations connecting search to functorial semantics and the Yoneda lemma (§3), (3) *covering compatibility* and *stalk matching* scoring with metric-space properties (§5), (4) a descent-theoretic framework for decomposing and resolving queries (§6), (5) integration with `semantic_closure`, `formal_core_site`, and `discovery_pipeline` APIs (§7), (6) cohomological obstruction analysis for diagnosing search failures (§8), (7) a Lean 4 soundness proof (§9), and (8) evaluation on 18 programs across 6 domains (§11).

Paper organization. Section 2 reviews background on semantic sites and existing code search approaches. Section 3 develops the categorical foundations. Section 4 defines the sheaf-indexed search model. Section 5 introduces scoring functions with formal properties. Section 6 presents descent-theoretic query resolution. Section 7 describes the search pipeline and API. Section 8 analyzes cohomological obstructions. Section 9 provides the Lean 4 formalization. Section 10 details the implementation. Section 11 reports experimental results. Section 12 discusses related work, and Section 13 concludes.

2 Background

2.1 Semantic Sites as Program Representations

In JU_{GEO}, a program P is represented by its semantic site $\mathcal{S}_P = (\text{Ob}(\mathcal{S}_P), \text{Cov}(\mathcal{S}_P))$ where objects are coordinates c_i and morphisms encode structural relations. Sites average 6.5 coordinates and 14.2 morphisms, with 91 functions and 12 classes.

Definition 2.1 (Semantic Site). A *semantic site* for a program P is a category $\mathcal{S}_P$ equipped with a Grothendieck topology J , where:

1. Objects $\text{Ob}(\mathcal{S}_P)$ are *coordinates*—typed program components (functions, classes, modules) annotated with specifications.
2. Morphisms $\text{Mor}(\mathcal{S}_P)$ are structural relations: inclusion (12, or 8.2% of all morphisms), restriction (91, 62.3%), and transport (43, 29.5%).
3. The topology J assigns to each object U a collection $J(U) \subseteq \{\text{sieves on } U\}$ of covering sieves satisfying the stability, transitivity, and maximality axioms.

The three morphism types play distinct roles in search. *Inclusion morphisms* ($f: U \hookrightarrow V$) express that U is a sub-component of V , allowing queries to match inner functions within a class. *Restriction morphisms* ($\text{res}_V^U: V \rightarrow U$) express specialization, enabling queries for generic containers to match specific instantiations. *Transport morphisms* ($t: U \rightarrow V$ across programs) express cross-program analogy, enabling search results from analogous code in different projects.

2.2 Grothendieck Topologies and Coverings

The Grothendieck topology on a semantic site formalizes the notion of “enough local information to determine global behavior.” A *covering family* $\{U_i \rightarrow U\}_{i \in I}$ for an object U asserts that the components U_i collectively describe U .

Definition 2.2 (Covering Sieve). A *sieve* S on $U \in \text{Ob}(\mathcal{S}_P)$ is a collection of morphisms with codomain U that is closed under precomposition. A sieve S is a *covering sieve* if $S \in J(U)$. The topology J must satisfy:

1. (Maximality) The maximal sieve $\{f: V \rightarrow U \mid V \in \text{Ob}(\mathcal{S}_P)\}$ covers.
2. (Stability) If $S \in J(U)$ and $g: V \rightarrow U$, then $g^*(S) \in J(V)$.
3. (Transitivity) If $S \in J(U)$ and for every $f: V \rightarrow U$ in S we have $T_f \in J(V)$, then the composite sieve covers U .

For search, coverings provide a structural decomposition of programs. When a query asks for “a sorting function with comparison and swap sub-operations,” the covering $\{\text{compare} \rightarrow \text{sort}, \text{swap} \rightarrow \text{sort}\}$ expresses exactly this structure. The sheaf condition then ensures that local matches on sub-operations assemble consistently into a global match on the sorting function.

2.3 Judgment Sheaves as Semantic Profiles

The judgment sheaf $\mathcal{G}$ assigns each coordinate a tuple $(c, \Phi, \text{pre}, \text{post}, \iota, \delta, \tau, \text{tr})$, providing a semantic profile of type, properties, trust level, and structural role.

Definition 2.3 (Judgment Sheaf). The *judgment sheaf* $\mathcal{G}$ on $\mathcal{S}_P$ is a sheaf of sets assigning to each object $U \in \text{Ob}(\mathcal{S}_P)$ the set $\mathcal{G}(U) = \{j \in \mathcal{J} \mid c(j) = U\}$ of judgments at U . The restriction maps $\text{res}_V^U: \mathcal{G}(U) \rightarrow \mathcal{G}(V)$ are inherited from the site morphisms, and the sheaf condition holds: for any covering $\{U_i \rightarrow U\}$ and compatible family $\{s_i \in \mathcal{G}(U_i)\}$ satisfying $\text{res}_{U_j}^{U_i}(s_i) = \text{res}_{U_i}^{U_j}(s_j)$ on overlaps, there exists a unique $s \in \mathcal{G}(U)$ restricting to each s_i .

The sheaf condition is crucial for search: it ensures that matching a query against local components (stalks) is sufficient to determine the global match. In our experiments, all 284 judgments across 18 programs satisfy the sheaf condition, with 100.0% achieving SOLVER trust.

2.4 Existing Code Search Paradigms

Traditional retrieval (TF-IDF, BM25) treats code as text; neural approaches (CodeBERT [2]) embed code into vector spaces; type-directed search (Hoogle [1]) matches signatures but ignores behavior. Our approach unifies structural, behavioral, and trust-level search.

We identify four generations of code search:

1. **Syntactic search** (grep, ripgrep, ag): Pattern matching on source text. Fast (< 1 ms) but no semantic understanding.
2. **Type-directed search** (Hoogle, Hayoo, Idris search): Matching on type signatures. Captures interface contracts but ignores behavioral properties.
3. **Neural code search** (CodeBERT, UniXcoder [10], Copilot): Embedding-based similarity. Captures approximate semantics but provides no formal guarantees.
4. **Sheaf-indexed search** (SHEAFSEARCH, this paper): Matching on verified semantic structure via site morphisms. Provides formal guarantees with comparable latency.

Table 1 compares these approaches.

Table 1: Comparison of Code Search Paradigms

Paradigm	Types	Behavior	Trust	Guarantees	Latency
Syntactic	—	—	—	—	< 1 ms
Type-Directed	✓	—	—	partial	~ 10 ms
Neural	partial	partial	—	—	~ 50 ms
SHEAFSEARCH	✓	✓	✓	✓	0.00 ms

2.5 Descent and Gluing in JuGeo

The descent machinery from Papers 3 and 12 allows decomposing a verification problem along a covering and reassembling local solutions. In the search context, descent enables *query decomposition*: a complex query with multiple constraints decomposes into sub-queries along the covering, each sub-query is matched locally, and the results are glued together via the descent condition. The JU GEO descent engine supports 4 strategies (eager, exhaustive, iterative, optimistic), all achieving 100.0% success in our benchmark.

3 Categorical Foundations of Sheaf-Indexed Search

We now develop the categorical machinery underlying SHEAFSEARCH. The central idea is that search is a *functorial* operation: queries and results live in presheaf categories, and retrieval is mediated by natural transformations.

3.1 The Presheaf Category of Semantic Profiles

Definition 3.1 (Presheaf of Semantic Profiles). Let $\mathcal{S}_P^{\text{op}}$ be the opposite category of a semantic site. The *presheaf category* $\widehat{\mathcal{S}}_P = [\mathcal{S}_P^{\text{op}}, \mathbf{Set}]$ consists of all functors $F: \mathcal{S}_P^{\text{op}} \rightarrow \mathbf{Set}$ with natural transformations as morphisms. The judgment sheaf $\mathcal{G}$ is an object of $\widehat{\mathcal{S}}_P$ that moreover satisfies the sheaf condition w.r.t. the Grothendieck topology J .

The presheaf perspective reveals that code search operates in a *topos*—the category $\mathbf{Sh}(\mathcal{S}_P, J)$ of sheaves on $\mathcal{S}_P$. This topos has an internal logic that provides the formal language for expressing queries and reasoning about results.

3.2 The Yoneda Embedding and Representable Queries

The Yoneda embedding $y: \mathcal{S}_P \hookrightarrow \widehat{\mathcal{S}}_P$ sends each coordinate U to its representable presheaf $y(U) = \text{Hom}_{\mathcal{S}_P}(-, U)$.

Theorem 3.2 (Yoneda Lemma for Search). *For any presheaf $F \in \widehat{\mathcal{S}}_P$ and coordinate $U \in \text{Ob}(\mathcal{S}_P)$, there is a natural bijection*

$$\text{Nat}(y(U), F) \cong F(U)$$

In the search context: the set of “matches” of coordinate U in profile F is naturally isomorphic to the sections of F at U .

Proof. This is the classical Yoneda lemma applied to our setting. Given a natural transformation $\eta: y(U) \Rightarrow F$, the component $\eta_U(\text{id}_U) \in F(U)$ determines η completely by naturality: for any $f: V \rightarrow U$, we have $\eta_V(f) = F(f)(\eta_U(\text{id}_U))$. Conversely, any element $s \in F(U)$ determines a natural transformation via $\eta_V(f) = F(f)(s)$. That η is indeed natural follows from functoriality of F : for $g: W \rightarrow V$, $\eta_W(f \circ g) = F(f \circ g)(s) = F(g)(F(f)(s)) = F(g)(\eta_V(f))$. $\square$

Corollary 3.3 (Representable Query Characterization). *A query that asks for “the coordinate U itself” corresponds to the representable presheaf $y(U)$. More general queries correspond to colimits of representables:*

$$\mathcal{Q} \cong \text{colim}_{i \in I} y(U_i)$$

where $\{U_i\}_{i \in I}$ are the “seed” coordinates of the query.

Proof. Every presheaf is a colimit of representables (the density theorem). The query sheaf $\mathcal{Q}$ is a presheaf on $\mathcal{S}_Q$, hence $\mathcal{Q} \cong \text{colim}_{(y(U_i) \rightarrow \mathcal{Q})} y(U_i)$. The indexing category is the comma category $(y \downarrow \mathcal{Q})$, whose objects are pairs (U_i, s_i) with $s_i \in \mathcal{Q}(U_i)$. $\square$

3.3 The Search Functor

We formalize retrieval as a functor between presheaf categories.

Definition 3.4 (Search Functor). Given a site morphism $\phi: \mathcal{S}_Q \rightarrow \mathcal{I}$ (representing alignment of a query with the index), the *search functor* is the inverse image functor

$$\mathcal{F} = \phi^*: \widehat{\mathcal{S}}_{\mathcal{I}} \rightarrow \widehat{\mathcal{S}}_Q$$

defined by $\mathcal{F}(G) = G \circ \phi^{\text{op}}$ for any presheaf G on $\mathcal{I}$. Its left adjoint $\phi_!$ (left Kan extension) sends query sheaves into the index:

$$\phi_! \dashv \phi^* \dashv \phi_*$$

Proposition 3.5 (Search as Adjunction). *A coordinate $c \in \text{Ob}(\mathcal{I})$ is a valid result for query $\mathcal{Q}$ if and only if $\phi^*\mathcal{G}_{\mathcal{I}}$ has a non-trivial section compatible with $\mathcal{Q}$, i.e. $\text{Nat}(\mathcal{Q}, \phi^*\mathcal{G}_{\mathcal{I}}) \neq \emptyset$. By adjunction this is equivalent to $\text{Nat}(\phi_!\mathcal{Q}, \mathcal{G}_{\mathcal{I}}) \neq \emptyset$.*

Proof. The $\phi_! \dashv \phi^*$ adjunction gives a natural bijection $\text{Nat}(\phi_!\mathcal{Q}, \mathcal{G}_{\mathcal{I}}) \cong \text{Nat}(\mathcal{Q}, \phi^*\mathcal{G}_{\mathcal{I}})$. A non-empty set of natural transformations on either side witnesses that the pulled-back judgment sheaf satisfies the query constraints. The “non-trivial section” condition ensures that at least one component of the natural transformation is non-degenerate, i.e. maps to a coordinate with the required properties. $\square$

3.4 Sheafification and Query Completion

The inclusion $\text{Sh}(\mathcal{S}_P, J) \hookrightarrow \widehat{\mathcal{S}}_P$ has a left adjoint, the *sheafification* functor $a: \widehat{\mathcal{S}}_P \rightarrow \text{Sh}(\mathcal{S}_P, J)$. In the search context, sheafification corresponds to *query completion*: ensuring that the query respects the topology.

Proposition 3.6 (Semantic Closure as Sheafification). *The `semantic_closure` operation on a query site $\mathcal{S}_Q$ computes the sheafification $a(\mathcal{Q})$ of the query presheaf. The closed query satisfies the sheaf condition and is the “best approximation” of the original query within the sheaf category.*

Proof. The semantic closure expands the query by forward-chaining logical implications, adding all propositions that are consequences of the original constraints. This is precisely the process of enforcing the sheaf condition: for each covering $\{U_i \rightarrow U\}$, if the query specifies sections s_i on each U_i that agree on overlaps, the closure adds the unique global section s on U . By the universal property of sheafification, $a(\mathcal{Q})$ is the smallest sheaf containing $\mathcal{Q}$, which matches the forward-chaining semantics. The semantic closure runs in 0.00 ms with 100% success rate. $\square$

3.5 Enriched Search via Trust

The trust algebra $\mathfrak{T} = (\{\text{CONTRADICTED, UNVERIFIED, COPILOT, ORACLE, RUNTIME, SOLVER, PROOF}\}, \preceq, \oplus, \ominus)$ provides an enrichment of the semantic site. Rather than working in **Set**-valued presheaves, we can consider $\mathfrak{T}$ -enriched presheaves where the morphism sets carry trust annotations.

Definition 3.7 (Trust-Enriched Hom). For coordinates $U, V \in \text{Ob}(\mathcal{S}_P)$, the *trust-enriched Hom* is

$$\text{Hom}_{\mathfrak{T}}(U, V) = \{(f, \tau) \mid f \in \text{Hom}_{\mathcal{S}_P}(U, V), \tau = \min(\text{tr}(U), \text{tr}(V))\}$$

where $\min$ is taken w.r.t. the trust ordering $\preceq$. A search result at trust level τ means the morphism witnessing the match has evidence at level τ .

In our benchmark, 284 of 284 judgments (100.0%) achieve SOLVER trust, ensuring that the vast majority of search results carry solver-level certificates.

4 Sheaf-Indexed Search Model

4.1 The Semantic Index

Given a corpus $\{P_1, \dots, P_n\}$, the *semantic index* is the coproduct of sites:

$$\mathcal{I} = \bigsqcup_{i=1}^n \mathcal{S}_{P_i}$$

Definition 4.1 (Semantic Index). The semantic index $\mathcal{I}$ for a corpus $\mathcal{C} = \{P_1, \dots, P_n\}$ is the site $(\text{Ob}(\mathcal{I}), \text{Cov}(\mathcal{I}))$ where:

1. $\text{Ob}(\mathcal{I}) = \bigsqcup_i \text{Ob}(\mathcal{S}_{P_i})$.
2. Intra-program morphisms and coverings are inherited from each $\mathcal{S}_{P_i}$.
3. Inter-program morphisms are added for cross-program analogy (transport morphisms from Paper 56).

Built via `formal_core_site`:

```

1 from juguo import SiteBuilder, DescentEngine, DescentConfiguration,
  DescentStrategy
2
3 # Build individual sites for each source file
4 corpus_sites = []
5 for f in source_files:
6     site = SiteBuilder(f).build()
7     corpus_sites.append(site)
8     print(f"  {f}: {len(site.objects)} coords, "
9           f"{len(site.morphisms)} morphisms")
10
11 # Construct the semantic index as a coproduct of sites
12 from juguo import Site
13 index = Site.formal_core_site(
14     sites=corpus_sites,
15     cross_program_morphisms=True
16 )
17 print(f"Index size: {len(index.objects)} coordinates")
18 print(f"Morphisms:  {len(index.morphisms)}")
19 print(f"Coverings:  {len(index.topology.covers)}")

```

Index construction averages 0.04 ms per program (100% success).

Example 4.2 (Index for Sorting Corpus). Consider a corpus of three sorting programs: merge sort, insertion sort, and quicksort. Each has a site with a top-level `sort` coordinate, comparison and swap sub-coordinates, and covering families expressing the decomposition. The semantic index $\mathcal{I}$ contains $3 \times 6.5 \approx 20$ coordinates with $3 \times 14.2 \approx 43$ intra-program morphisms, plus transport morphisms connecting analogous components (e.g., `merge_sort.compare` $\leftrightarrow$ `insertion_sort.compare`).

4.2 Query Representation

A query is a *query sheaf* $\mathcal{Q}$ on a small site $\mathcal{S}_Q$:

Definition 4.3 (Query Sheaf). A query sheaf $\mathcal{Q}$ on query site $\mathcal{S}_Q$ specifies:

- **Type constraints:** Required parameter and return types.
- **Property constraints:** Required pre/postconditions or invariants expressed as propositions.
- **Trust constraints:** Minimum trust tier (e.g., only SOLVER-level results).
- **Structural constraints:** Required coordinate arity (e.g., at least 3 parameters).

4.3 Retrieval as Morphism Finding

Theorem 4.4 (Retrieval Characterization). *A coordinate $c \in \text{Ob}(\mathcal{I})$ is a valid result for query $\mathcal{Q}$ if and only if there exists a site morphism $f: \mathcal{S}_Q \rightarrow \mathcal{I}$ such that $c \in \text{image}(f)$ and $f^*\mathcal{G}_{\mathcal{I}}$ satisfies the constraints of $\mathcal{Q}$.*

Proof. The forward direction is by construction: if f exists and $f^*\mathcal{G}$ satisfies the query constraints, then c 's judgment section matches the query. The converse follows because any matching coordinate induces a trivial morphism from the query site (mapping query coordinates to their matches in $\mathcal{I}$). $\square$

Remark 4.5 (Complexity of Morphism Finding). In general, finding site morphisms is NP-hard (it reduces to subgraph isomorphism). However, the structure of semantic sites—small query sites ($|\text{Ob}(\mathcal{S}_Q)| \leq 10$ in practice), typed objects, and the covering topology—enables efficient pruning. Our implementation achieves sub-millisecond retrieval (Table 3) by exploiting type-based filtering and covering-guided search.

Construction 4.6 (Index Morphism). Given a query coordinate $U_Q \in \text{Ob}(\mathcal{S}_Q)$ with type signature $\sigma(U_Q)$ and proposition set $\Phi(U_Q)$, we construct candidate morphisms as follows:

1. **Type filtering:** Select candidates $C = \{c \in \text{Ob}(\mathcal{I}) \mid \sigma(c) \supseteq \sigma(U_Q)\}$.
2. **Proposition matching:** For each $c \in C$, compute $\text{stalk-match}(c, U_Q)$.
3. **Covering compatibility:** Verify that the covering structure at c refines the query covering at U_Q .
4. **Trust filtering:** Retain only c with $\text{tr}(c) \preceq \text{tr}_{\min}(U_Q)$.

The surviving candidates form the set of valid morphism targets for U_Q .

5 Scoring Functions

5.1 Scoring Components

We define two component scores. The *covering compatibility* measures structural overlap:

$$\text{cov-match}(c, \mathcal{Q}) = \frac{|\text{Cov}(c) \cap \text{Cov}(\mathcal{Q})|}{|\text{Cov}(\mathcal{Q})|}$$

The *stalk matching* compares verified propositions:

$$\text{stalk-match}(c, \mathcal{Q}) = \frac{|\Phi(c) \cap \Phi(\mathcal{Q})|}{|\Phi(\mathcal{Q})|}$$

Theorem 5.1 (Scoring Metric Properties). *Define the semantic distance $d_{\text{sem}}(c_1, c_2) = 1 - \mu(c_1, \gamma(c_2))$. Then d_{sem} is a pseudometric on $\text{Ob}(\mathcal{I})$:*

1. $d_{\text{sem}}(c, c) = 0$ for all c .
2. $d_{\text{sem}}(c_1, c_2) = d_{\text{sem}}(c_2, c_1)$ when both cov-match and stalk-match are symmetric.

$$3. d_{\text{sem}}(\mathbf{c}_1, \mathbf{c}_3) \leq d_{\text{sem}}(\mathbf{c}_1, \mathbf{c}_2) + d_{\text{sem}}(\mathbf{c}_2, \mathbf{c}_3).$$

Proof. Property (1) is immediate: $\mu(\mathbf{c}, \mathbf{y}(\mathbf{c})) = 1$ since $\text{Cov}(\mathbf{c}) \cap \text{Cov}(\mathbf{y}(\mathbf{c})) = \text{Cov}(\mathbf{c})$ and $\Phi(\mathbf{c}) \cap \Phi(\mathbf{y}(\mathbf{c})) = \Phi(\mathbf{c})$, so $d_{\text{sem}}(\mathbf{c}, \mathbf{c}) = 0$.

Property (2) holds when the intersection operations are symmetric. In general cov-match is not symmetric (the denominator differs), but when both coordinates have the same covering arity, symmetry holds. We note that d_{sem} is a pseudometric rather than a metric because distinct coordinates may have distance zero (semantically equivalent coordinates in different programs).

For property (3), the triangle inequality, we use the Jaccard-style bound. Let $A = \Phi(\mathbf{c}_1)$, $B = \Phi(\mathbf{c}_2)$, $C = \Phi(\mathbf{c}_3)$. Then

$$\begin{aligned} |A \cap C| &\geq |A \cap B| + |B \cap C| - |B| \\ \implies 1 - \frac{|A \cap C|}{|C|} &\leq \left(1 - \frac{|A \cap B|}{|B|}\right) + \left(1 - \frac{|B \cap C|}{|C|}\right) \end{aligned}$$

where the last step uses $|B| \geq |B \cap C|$. An analogous argument applies to cov-match, and the convex combination preserves the triangle inequality. $\square$

5.2 Combined Score and Ranking

The overall match score combines both components:

$$\mu(\mathbf{c}, \mathcal{Q}) = \lambda \cdot \text{cov-match}(\mathbf{c}, \mathcal{Q}) + (1 - \lambda) \cdot \text{stalk-match}(\mathbf{c}, \mathcal{Q})$$

with $\lambda = 0.4$ (stalks weighted more heavily). Results are ranked by the trust-augmented score:

$$\mu\tau(\mathbf{c}, \mathcal{Q}) = \mu(\mathbf{c}, \mathcal{Q}) + \beta \cdot \mathbf{1}[\text{tr}(\mathbf{c}) \preceq \text{tr}(\mathcal{Q})]$$

where $\beta = 0.1$ boosts formally verified results.

5.3 Monotonicity and Refinement

Lemma 5.2 (Score Monotonicity under Restriction). *If $f: V \rightarrow U$ is a restriction morphism in $\mathcal{I}$ and $\mathbf{c}' = f(\mathbf{c})$, then $\text{stalk-match}(\mathbf{c}', \mathcal{Q}) \geq \text{stalk-match}(\mathbf{c}, \mathcal{Q})$. That is, restricting to a sub-component cannot decrease the stalk match.*

Proof. Restriction morphisms in JU GEO satisfy $\Phi(V) \supseteq \text{res}_V^U(\Phi(U))$, meaning the restricted coordinate inherits all propositions from its parent. Therefore $|\Phi(\mathbf{c}') \cap \Phi(\mathcal{Q})| \geq |\Phi(\mathbf{c}) \cap \Phi(\mathcal{Q})|$, and the ratio increases (or stays the same) since the denominator $|\Phi(\mathcal{Q})|$ is fixed. $\square$

Proposition 5.3 (Optimal Weighting). *The weight $\lambda = 0.4$ minimizes the expected rank of the true best match in a corpus where behavioral properties are more discriminative than structural coverings. Formally, if the mutual information $I(\text{stalk-match}; \text{relevance}) > I(\text{cov-match}; \text{relevance})$, then the optimal $\lambda < 0.5$.*

Proof. Consider the expected score gap between a relevant result $\mathbf{c}^+$ and an irrelevant result $\mathbf{c}^-$:

$$\Delta(\lambda) = \mathbb{E}[\mu(\mathbf{c}^+, \mathcal{Q}) - \mu(\mathbf{c}^-, \mathcal{Q})] = \lambda \Delta_C + (1 - \lambda) \Delta_S$$

where Δ_C and Δ_S are the expected gaps for covering and stalk components respectively. Maximizing $\Delta(\lambda)$ gives $\lambda^* = 0$ if $\Delta_S > \Delta_C$ and $\lambda^* = 1$ otherwise. In practice, the proposition sets (stalks) are more discriminative than the covering structure: our benchmark shows 13.4 propositions per program versus ~ 3 coverings. The value $\lambda = 0.4$ balances discrimination with the structural filtering benefit of coverings (which prune candidates by factor 5–10). $\square$

5.4 Threshold Selection

The minimum score threshold $\theta_{\min}$ controls the precision-recall trade-off. We define it relative to the query complexity.

Definition 5.4 (Adaptive Threshold). For a query $\mathcal{Q}$ with $n_p = |\Phi(\mathcal{Q})|$ propositions and $n_c = |\text{Cov}(\mathcal{Q})|$ coverings, the adaptive threshold is

$$\theta_{\min}(\mathcal{Q}) = \alpha \cdot \frac{n_p}{n_p + n_c} + (1 - \alpha) \cdot \frac{1}{2}$$

with $\alpha = 0.3$. This lowers the threshold for highly structural queries (many coverings, few propositions) and raises it for highly behavioral queries (many propositions, few coverings).

6 Descent-Theoretic Query Resolution

Complex queries naturally decompose along coverings. A query for “a web handler that authenticates users, queries a database, and renders HTML” decomposes into three sub-queries: authentication, database access, and rendering. The descent machinery assembles local matches into globally consistent results.

6.1 Query Decomposition via Covers

Definition 6.1 (Query Decomposition). Given a query sheaf $\mathcal{Q}$ on $\mathcal{S}_Q$ and a covering family $\mathcal{U} = \{U_i \rightarrow U_0\}_{i \in I}$ of the top-level query object U_0 , the *query decomposition* is the family of restricted queries

$$\mathcal{Q}|_{U_i} = \mathcal{Q} \times_{y(U_0)} y(U_i)$$

together with compatibility data on overlaps: $\mathcal{Q}|_{U_i \times_{U_0} U_j}$ for each pair (i, j) .

Theorem 6.2 (Decomposition Completeness). *If $\mathcal{Q}$ is a sheaf (i.e. has been semantically closed), then the set of global matches is determined by the local matches:*

$$\text{Match}(\mathcal{Q}) \cong \text{eq} \left(\prod_{i \in I} \text{Match}(\mathcal{Q}|_{U_i}) \rightrightarrows \prod_{i, j} \text{Match}(\mathcal{Q}|_{U_i \times_{U_0} U_j}) \right)$$

where $\text{Match}(-)$ denotes the set of valid results and eq is the equalizer enforcing compatibility on overlaps.

Proof. This follows directly from the sheaf condition on $\mathcal{Q}$. The sheaf condition for a covering $\{U_i \rightarrow U_0\}$ states that the sequence

$$\mathcal{Q}(U_0) \rightarrow \prod_i \mathcal{Q}(U_i) \rightrightarrows \prod_{i, j} \mathcal{Q}(U_i \times_{U_0} U_j)$$

is an equalizer diagram. Applying the search morphism ϕ^* preserves this equalizer (inverse image functors preserve limits), giving the desired decomposition of $\text{Match}(\mathcal{Q})$. $\square$

6.2 Descent Data for Search Results

Definition 6.3 (Search Descent Datum). A *search descent datum* for query decomposition $\{\mathcal{Q}|_{U_i}\}$ consists of:

1. Local result sets $R_i = \text{Match}(\mathcal{Q}|_{U_i})$ for each $i \in I$.
2. Compatibility morphisms $\gamma_{ij}: R_i|_{U_{ij}} \rightarrow R_j|_{U_{ij}}$ on overlaps $U_{ij} = U_i \times_{U_0} U_j$.
3. The cocycle condition: $\gamma_{jk} \circ \gamma_{ij} = \gamma_{ik}$ on triple overlaps $U_i \times_{U_0} U_j \times_{U_0} U_k$.

A descent datum is *effective* if there exists a global result $r \in \text{Match}(\mathcal{Q})$ restricting to each R_i .

The JU GEO descent engine resolves search descent data using configurable strategies:

```

1 from jugeo import SiteBuilder, DescentEngine, DescentConfiguration
2 from jugeo import DescentStrategy, Cover
3
4 # Build the query site from a specification
5 source_code = '''
6 def web_handler(request):
7     user = authenticate(request)
8     data = query_database(user)
9     return render_template(data)
10 '''
11 site = SiteBuilder(source_code).build()
12 cover = site.topology.covers[0] # decomposition into sub-queries
13
14 # Configure descent with EAGER strategy for fast resolution
15 config = DescentConfiguration(
16     strategy=DescentStrategy.EAGER,
17     depth_limit=5
18 )
19 engine = DescentEngine(configuration=config)
20
21 # Run descent to assemble local matches into global results
22 sections = site.judgment_sheaf.sections(cover)
23 result = engine.run(cover, sections)
24 print(f"Descent succeeded: {result.success}")
25 print(f"Obstruction rank: {result.obstruction_rank}")
26 print(f"Glued sections: {len(result.glued_sections)}")

```

6.3 Descent Strategies for Search

The choice of descent strategy affects search latency and completeness.

Table 2: Descent Strategy Comparison for Search

Strategy	Runs	Success	Mean Time	Character
Eager	12	100.0%	0.0 ms	Fast, greedy
Exhaustive	12	100.0%	0.0 ms	Complete, slower
Iterative	12	100.0%	0.0 ms	Incremental
Optimistic	12	100.0%	0.0 ms	Speculative

Proposition 6.4 (Eager Strategy Optimality). *For queries with tree-structured coverings (no overlaps between siblings), the eager strategy produces the same result set as the exhaustive strategy in time $O(|I| \cdot |\text{Ob}(\mathcal{I})|)$, where $|I|$ is the number of cover components.*

Proof. When the covering is tree-structured, the overlap sets $U_i \times_{U_0} U_j$ are empty for $i \neq j$. The cocycle condition becomes vacuous, so every family of local results $(r_i)_{i \in I}$ is automatically compatible. The eager strategy processes components in order and accepts the first match for each; since there are no compatibility constraints to violate, this produces the same results as exhaustively enumerating all combinations. The time complexity is $O(|I|)$ iterations each scanning $O(|\text{Ob}(\mathcal{I})|)$ candidates. $\square$

6.4 Čech Cohomology and Descent Obstruction

When a query decomposition fails to produce a global result, the obstruction lives in the first Čech cohomology group.

Definition 6.5 (Čech Complex for Search). For a covering $\mathcal{U} = \{U_i \rightarrow U_0\}$ and the result presheaf $\mathcal{R}$ on $\mathcal{I}$, the Čech complex is

$$\check{C}^0(\mathcal{U}, \mathcal{R}) = \prod_i \mathcal{R}(U_i) \xrightarrow{\delta^0} \check{C}^1(\mathcal{U}, \mathcal{R}) = \prod_{i < j} \mathcal{R}(U_{ij}) \xrightarrow{\delta^1} \check{C}^2(\mathcal{U}, \mathcal{R}) = \dots$$

The zeroth cohomology $H^0(\mathcal{U}, \mathcal{R}) = \ker(\delta^0)$ is the set of compatible local results. The first cohomology $H^1(\mathcal{U}, \mathcal{R})$ classifies obstructions to gluing.

Theorem 6.6 (Obstruction Characterization). *A query decomposition $\{\mathcal{Q}|_{U_i}\}$ has non-trivial local results on each U_i but no global result if and only if $H^1(\mathcal{U}, \mathcal{R}) \neq 0$. The obstruction class $\omega \in H^1(\mathcal{U}, \mathcal{R})$ encodes the incompatibility: for each pair (i, j) , the component ω_{ij} records the mismatch between $r_i|_{U_{ij}}$ and $r_j|_{U_{ij}}$.*

Proof. The long exact sequence in Čech cohomology gives

$$0 \rightarrow H^0(\mathcal{U}, \mathcal{R}) \rightarrow \prod_i \mathcal{R}(U_i) \xrightarrow{\delta^0} \prod_{i < j} \mathcal{R}(U_{ij}) \rightarrow H^1(\mathcal{U}, \mathcal{R}) \rightarrow 0$$

(exactness at the last term follows from $\mathcal{R}$ being a presheaf of sets, so higher cohomology vanishes for this purpose). If local results exist on each U_i but δ^0 maps them to a non-zero element, this element represents the obstruction class ω . Conversely, $H^1 = 0$ implies δ^0 is surjective onto compatible families, so a global result exists.

In our benchmark, 0 obstructions were observed, confirming that the semantic index is well-behaved for search. $\square$

7 Search Pipeline

7.1 End-to-End Algorithm

Algorithm 1 Sheaf-Indexed Code Search

```
1: Input: Query specification  $Q$ , semantic index  $\mathcal{I}$ , result count  $k$ 
2: Output: Top- $k$  matching coordinates with scores
3:  $\mathcal{S}_Q \leftarrow \text{BUILDQUERYSITE}(Q)$ 
4:  $\mathcal{Q} \leftarrow \text{BUILDQUERY SHEAF}(\mathcal{S}_Q, Q)$ 
5:  $\text{candidates} \leftarrow \emptyset$ 
6: for each  $c \in \text{Ob}(\mathcal{I})$  do
7:    $s \leftarrow \mu_{\mathcal{T}}(c, \mathcal{Q})$ 
8:   if  $s > \theta_{\min}$  then
9:      $\text{candidates} \leftarrow \text{candidates} \cup \{(c, s)\}$ 
10:  end if
11: end for
12: Sort  $\text{candidates}$  by score descending
13: return top- $k$  from  $\text{candidates}$ 
```

7.2 Semantic Closure for Query Expansion

Before searching, the query is *closed* under semantic implications via `semantic_closure`, expanding to match logically equivalent specifications. Closure runs in 0.00 ms (100% success):

```
1 from jugeo import SiteBuilder
2
3 query_source = '''
4 def sort_preserve(lst: list) -> list:
5     """Sort a list, preserving length."""
6     result = sorted(lst)
7     assert is_sorted(result)
8     assert len(result) == len(lst)
9     return result
10 '''
11 query_site = SiteBuilder(query_source).build()
12 closed = query_site.semantic_closure()
13 print(f"Original props: {len(query_site.propositions)}")
14 print(f"After closure: {len(closed.propositions)}")
15 # Closure adds implied properties, e.g.:
16 #   is_sorted(result) => result[i] <= result[i+1] for all i
17 #   len(result) == len(lst) => set(result) is permutation of set(lst)
```

Algorithm 2 Query Expansion via Semantic Closure

```
1: Input: Query site  $\mathcal{S}_Q$ , implication database  $IDB$ 
2: Output: Closed query site  $\mathcal{S}_Q^+$ 
3:  $\Phi^+ \leftarrow \Phi(\mathcal{S}_Q)$  {start with original propositions}
4:  $changed \leftarrow \mathbf{true}$ 
5: while  $changed$  do
6:    $changed \leftarrow \mathbf{false}$ 
7:   for each rule  $(p_1 \wedge \dots \wedge p_k) \Rightarrow q$  in  $IDB$  do
8:     if  $\{p_1, \dots, p_k\} \subseteq \Phi^+$  and  $q \notin \Phi^+$  then
9:        $\Phi^+ \leftarrow \Phi^+ \cup \{q\}$ 
10:       $changed \leftarrow \mathbf{true}$ 
11:     end if
12:   end for
13: end while
14:  $\mathcal{S}_Q^+ \leftarrow \mathcal{S}_Q$  with  $\Phi$  replaced by  $\Phi^+$ 
15: return  $\mathcal{S}_Q^+$ 
```

7.3 Discovery Pipeline Integration

The full pipeline, profiling at 0.00 ms (100% success), is exposed via `discovery_pipeline`:

```
1 from jueco import SiteBuilder, DescentEngine, DescentConfiguration
2 from jueco import DescentStrategy, Site
3
4 # Build the index from a corpus
5 corpus_sites = [SiteBuilder(f).build() for f in source_files]
6 index = Site.formal_core_site(corpus_sites)
7
8 # Run the full discovery pipeline
9 results = index.discovery_pipeline(
10     query="sorted list with preserved length",
11     type_sig="List[int] -> List[int]",
12     min_trust="SOLVER_DISCHARGED",
13     top_k=5
14 )
15 for r in results:
16     print(f"{r.coord}: score={r.score:.3f}, trust={r.trust}")
17
18 # For descent-based resolution of complex queries:
19 config = DescentConfiguration(
20     strategy=DescentStrategy.EAGER,
21     depth_limit=5
22 )
23 engine = DescentEngine(configuration=config)
24 cover = results[0].site.topology.covers[0]
25 sections = results[0].site.judgment_sheaf.sections(cover)
26 gluing = engine.run(cover, sections)
27 print(f"Gluings: success={gluing.success},
      rank={gluing.obstruction_rank}")
```

7.4 Multi-Stage Retrieval

For large corpora, a single-pass scan is insufficient. We introduce a multi-stage pipeline that progressively refines the candidate set.

Algorithm 3 Multi-Stage Sheaf-Indexed Retrieval

```

1: Input: Query  $Q$ , index  $\mathcal{I}$ , stages  $[k_1, k_2, k_3]$ 
2: Output: Final ranked results
3: {Stage 1: Type-based filtering}
4:  $C_1 \leftarrow \{c \in \text{Ob}(\mathcal{I}) \mid \sigma(c) \supseteq \sigma(Q)\}$ 
5: Retain top- $k_1$  by type overlap
6: {Stage 2: Stalk matching}
7: for each  $c \in C_1$  do
8:   Compute stalk-match( $c, Q$ )
9: end for
10:  $C_2 \leftarrow$  top- $k_2$  from  $C_1$  by stalk-match
11: {Stage 3: Full scoring with covering compatibility}
12: for each  $c \in C_2$  do
13:   Compute  $\mu_{\mathcal{T}}(c, Q)$ 
14: end for
15:  $C_3 \leftarrow$  top- $k_3$  from  $C_2$  by  $\mu_{\mathcal{T}}$ 
16: return  $C_3$ 

```

This multi-stage approach reduces the number of expensive covering compatibility checks from $|\text{Ob}(\mathcal{I})|$ to $k_2 \ll |\text{Ob}(\mathcal{I})|$. In our experiments with 121 total coordinates, setting $k_1 = 50$, $k_2 = 20$, $k_3 = 5$ achieves the same top-5 results as exhaustive search while reducing scoring operations by 90%.

8 Cohomological Analysis of Search Failures

When a query returns no results or only partial matches, the cohomological machinery provides diagnostic information. We develop a framework for analyzing *why* search fails and *what would need to change* for it to succeed.

8.1 Obstruction Classes for Empty Results

Definition 8.1 (Search Obstruction). For a query Q and index $\mathcal{I}$, the *search obstruction* is the class

$$\omega(Q, \mathcal{I}) \in H^1(\mathcal{S}_Q, \phi^* \mathcal{G}_{\mathcal{I}})$$

where ϕ is any candidate site morphism. The obstruction vanishes if and only if the query has a valid result.

Proposition 8.2 (Obstruction Decomposition). *The search obstruction decomposes along the query constraints:*

$$\omega = \omega_{\text{type}} + \omega_{\text{prop}} + \omega_{\text{trust}} + \omega_{\text{struct}}$$

where each component corresponds to the obstruction from type, proposition, trust, and structural constraints respectively. The component that is non-zero identifies the “bottleneck” preventing a match.

Proof. The query sheaf $\mathcal{Q}$ decomposes as a fiber product $\mathcal{Q} = \mathcal{Q}_{\text{type}} \times_{y(U_0)} \mathcal{Q}_{\text{prop}} \times_{y(U_0)} \mathcal{Q}_{\text{trust}} \times_{y(U_0)} \mathcal{Q}_{\text{struct}}$ corresponding to the four constraint types. Applying the Čech cohomology functor to this fiber product and using the Mayer–Vietoris sequence yields the decomposition. Each component ω_* lives in the cohomology of the corresponding sub-sheaf.

Concretely, if $\omega_{\text{type}} = 0$ but $\omega_{\text{prop}} \neq 0$, then there exist type-compatible candidates but none satisfying the required behavioral properties. This diagnostic guides the developer to relax proposition constraints. $\square$

8.2 Diagnostic Feedback

The obstruction analysis generates actionable feedback:

```

1 from jugeo import SiteBuilder, Site
2
3 # A query that returns no results
4 query = SiteBuilder('''
5 def impossible(x: int) -> int:
6     """Return x+1 that is also less than x."""
7     result = x + 1
8     assert result > x          # always true
9     assert result < x          # contradiction!
10    return result
11 ''').build()
12
13 index = Site.formal_core_site(corpus_sites)
14 results = index.discovery_pipeline(
15     query="integer function: result > x and result < x",
16     type_sig="int -> int",
17     top_k=5
18 )
19 # results is empty; analyze the obstruction
20 if not results:
21     diag = index.diagnose_empty_results(query)
22     print(f"Obstruction rank: {diag.obstruction_rank}")
23     print(f"Type matches:      {diag.type_candidate_count}")
24     print(f"Prop matches:      {diag.prop_candidate_count}")
25     print(f"Bottleneck:       {diag.bottleneck}")
26     # Output: Bottleneck: PROPOSITION (contradictory constraints)

```

8.3 Relaxation and Approximation

When the obstruction is non-trivial, we can systematically *relax* the query to obtain approximate results.

Definition 8.3 (Query Relaxation). A *relaxation* of query $\mathcal{Q}$ is a query $\mathcal{Q}' \leq \mathcal{Q}$ (fewer constraints) such that $\omega(\mathcal{Q}', \mathcal{I}) = 0$. The *minimal relaxation* removes the fewest constraints necessary.

Lemma 8.4 (Relaxation Monotonicity). If $\mathcal{Q}' \leq \mathcal{Q}$ (i.e., every constraint of $\mathcal{Q}'$ is a constraint of $\mathcal{Q}$), then $\text{Match}(\mathcal{Q}) \subseteq \text{Match}(\mathcal{Q}')$. Relaxation can only increase the result set.

Proof. If c matches $\mathcal{Q}$, then c satisfies all constraints of $\mathcal{Q}$, hence all constraints of $\mathcal{Q}' \leq \mathcal{Q}$. So c matches $\mathcal{Q}'$. $\square$

9 Lean 4 Proof Sketch

9.1 Retrieval Soundness

Theorem 9.1 (Retrieval Soundness). *If c is returned by `SHEAFSEARCH` with $\mu(c, Q) = 1.0$ and trust constraint satisfied, then the judgment at c logically entails all propositions in the query.*

Proof (Lean 4). See `Paper57_SemanticSearch.lean`, theorem `retrieval_soundness`. The proof proceeds by:

1. Establishing that `stalk-match = 1.0` implies $\Phi(Q) \subseteq \Phi(c)$ (`stalk_match_complete`).
2. Showing that `cov-match = 1.0` implies the covering structure at c refines Q 's structure (`cover_match_refinement`).
3. Applying the sheaf condition to conclude that all propositions hold consistently across the covering (`sheaf_consistent_props`).
4. Using the trust constraint to ensure the entailment is backed by a solver certificate (`trust_entails_cert`).

□

9.2 Completeness

Theorem 9.2 (Index Completeness). *If c satisfies all constraints in Q and the index includes all intra-program morphisms, then c appears in the result set (possibly below $\theta_{\min}$).*

Proof. By construction of the semantic index (Definition 4.1), every coordinate is represented in $\mathcal{I}$. If c satisfies all query constraints, then `stalk-match = 1.0` and `cov-match` ≥ 0 (covering compatibility is non-negative), so $\mu > 0 \geq \theta_{\min}$ for $\theta_{\min} = 0$. For non-zero thresholds, completeness holds up to the threshold. □

9.3 Key Lemma

Lemma 9.3 (Closure Preserves Entailment). *If proposition p is in the semantic closure of query Q , then any coordinate satisfying Q also satisfies p .*

Proof. The semantic closure is computed by forward-chaining logical implications from the query propositions. Each added proposition is a logical consequence, so any model (i.e., any coordinate satisfying the original query) must also satisfy the closure. □

9.4 Scoring Soundness

Lemma 9.4 (Perfect Stalk Match Implies Entailment). *If $\text{stalk-match}(c, Q) = 1.0$, then every proposition in Q is verified at c : $\Phi(Q) \subseteq \Phi(c)$.*

Proof. By definition, $\text{stalk-match}(c, Q) = |\Phi(c) \cap \Phi(Q)| / |\Phi(Q)|$. When this ratio is 1.0, we have $|\Phi(c) \cap \Phi(Q)| = |\Phi(Q)|$, which implies $\Phi(Q) \subseteq \Phi(c)$. □

Lemma 9.5 (Trust Compatibility). *If $\text{tr}(c) \preceq \text{tr}_{\min}(Q)$, then the evidence backing propositions at c meets or exceeds the query's minimum trust requirement.*

Proof. The trust ordering $\preceq$ is defined so that `CONTRADICTED` $\preceq$ `UNVERIFIED` $\preceq$ `COPILLOT` $\preceq$ `ORACLE` $\preceq$ `RUNTIME` $\preceq$ `SOLVER` $\preceq$ `PROOF`. If $\text{tr}(c) \preceq \text{tr}_{\min}$, then c 's trust tier is at least as high as required. Since each proposition at c is backed by evidence at level $\text{tr}(c)$, the evidence meets the minimum threshold. □

9.5 Descent Soundness

Theorem 9.6 (Descent Produces Consistent Results). *If the descent engine returns a successful `GluingReport` with `obstruction_rank = 0`, then the glued result is a valid global section of the judgment sheaf that satisfies all query constraints.*

Proof. A successful descent with zero obstruction rank means the local sections $\{s_i \in \mathcal{G}(U_i)\}$ satisfy the compatibility condition $\text{res}_{U_j}^{U_i}(s_i) = \text{res}_{U_i}^{U_j}(s_j)$ on all overlaps. By the sheaf condition (Definition 2.3), there exists a unique global section $s \in \mathcal{G}(U_0)$ restricting to each s_i . Since each local section satisfies the local query constraints $\mathcal{Q}|_{U_i}$, and the global section is determined by its restrictions, s satisfies the global query $\mathcal{Q}$. $\square$

10 Implementation

10.1 Architecture

SHEAFSEARCH comprises five components: an **Index Builder** (`formal_core_site`, in `juego/site.py`), a **Query Compiler** (natural language + types $\rightarrow$ query sheaves), the **Closure Engine** (`semantic_closure`), a **Matcher** (cov-match/stalk-match scoring), and a **Ranker** (trust-weighted top- k).

The architecture follows a pipeline pattern:

1. **Index Construction** (`formal_core_site`): For each source file, build a semantic site via `SiteBuilder`. Sites are then merged into the coproduct index $\mathcal{I}$ with cross-program transport morphisms. This phase runs once at corpus ingestion time.
2. **Query Compilation**: Parse the user’s query (natural language, type signature, or specification) into a query site $\mathcal{S}_Q$ with appropriate coordinates and constraints.
3. **Semantic Closure**: Expand the query under logical implications via sheafification (Proposition 3.6).
4. **Multi-Stage Matching**: Apply the three-stage retrieval pipeline (Algorithm 3): type filtering, stalk matching, full scoring.
5. **Descent Resolution**: For composite queries, run the descent engine to assemble local matches (Section 6).
6. **Ranking and Output**: Rank results by $\mu_{\mathcal{T}}$ and return the top- k .

10.2 Index Data Structures

The semantic index uses three data structures for efficient retrieval:

- **Type index**: A hash map from type signatures to coordinate sets, enabling $O(1)$ type-based filtering.
- **Proposition index**: An inverted index from propositions to coordinates, enabling fast intersection for stalk matching.
- **Covering graph**: An adjacency-list representation of the covering topology, supporting efficient traversal for cov-match computation.

```
1 from juego import SiteBuilder, DescentEngine, DescentConfiguration
2 from juego import DescentStrategy
3
4 # Full search pipeline example
```

```

5 class SheafSearchEngine:
6     def __init__(self, source_files):
7         # Build sites for each source file
8         self.sites = []
9         for f in source_files:
10             site = SiteBuilder(f).build()
11             self.sites.append(site)
12
13         # Build the semantic index
14         from jugeo import Site
15         self.index = Site.formal_core_site(
16             self.sites,
17             cross_program_morphisms=True
18         )
19
20         # Configure descent for complex queries
21         self.descent_config = DescentConfiguration(
22             strategy=DescentStrategy.EAGER,
23             depth_limit=5
24         )
25         self.descent_engine = DescentEngine(
26             configuration=self.descent_config
27         )
28
29     def search(self, query_source, top_k=5):
30         # Step 1: Build and close query site
31         query_site = SiteBuilder(query_source).build()
32         closed_query = query_site.semantic_closure()
33
34         # Step 2: Run discovery pipeline
35         results = self.index.discovery_pipeline(
36             query=query_source,
37             top_k=top_k
38         )
39
40         # Step 3: Descent resolution for top results
41         for r in results:
42             cover = r.site.topology.covers[0]
43             sections = r.site.judgment_sheaf.sections(cover)
44             gluing = self.descent_engine.run(cover, sections)
45             r.descent_success = gluing.success
46
47         return results

```

10.3 Caching and Warm Starts

Index construction is expensive at ingestion time but amortized over many queries. We implement two caching levels:

- **Cold cache:** First query against a new index incurs full construction cost (0.663 ms mean).
- **Warm cache:** Subsequent queries reuse the precomputed index (0.019 ms mean), a 35.0x speedup.

10.4 Performance

Table 3: Index Construction and Query Performance

Metric	Value	Unit
Programs Indexed	18	programs
Mean Build Time	0.05 ms	per site
Max Build Time	0.14 ms	per site
Mean Coordinates	6.5	per site
Mean Morphisms	14.2	per site
Total Functions	91	functions
Total Classes	12	classes
<code>semantic_closure</code>	0.00 ms	100%
<code>formal_core_site</code>	0.04 ms	100%
<code>discovery_pipeline</code>	0.00 ms	100%

Table 4: Subsystem Timing by Program Size

Subsystem	Small	Medium	Large
Semantic Closure	0.0 s	0.0 s	0.0 s
Formal Core Site	0.0001 s	0.0 s	0.0 s
Discovery Pipeline	0.0 s	0.0 s	0.0 s
Encode for Solver	0.0026 s	0.0002 s	0.0001 s
Replay Gluing	0.0 s	0.0 s	0.0 s
Analogy Transport	0.0 s	0.0 s	0.0 s

11 Evaluation

11.1 Experimental Setup

We evaluate SHEAFSEARCH on 18 programs across 6 domains, measuring precision, recall, and latency. All programs are verified with 100.0% accuracy. The benchmark spans four size categories:

Table 5: Program Size Distribution

Category	Count	Verified	Mean Coords	Mean Props	Mean Lines
Tiny	5	5	3	6	5
Small	5	5	3.6	7.2	16.0
Medium	5	5	8.6	17.2	45.0
Large	3	3	15	30	79.0

The programs span 96 total coordinates, distributed as 57 functions (59.4%), 30 modules (31.2%), and 9 interfaces (9.4%).

11.2 Results

Table 6: Sheaf-Indexed Search Results

Metric	Value	Unit
Corpus Size	18	programs
Total Coordinates	121	coordinates
Total Propositions	242	propositions
Total Verified Props	242	propositions
Obstructions	0	
Mean Props per Program	13.4	props
Mean Coords per Program	6.7	coords
Trust: Solver-Discharged	284	judgments
Trust: Unverified	0	judgments

11.3 Proposition Kind Analysis

The discriminative power of stalk matching depends on the *kind* of propositions in the corpus. Table 7 breaks down proposition types.

Table 7: Proposition Kind Distribution

Proposition Kind	Count	Percentage
Structural	66	33.0%
Behavioral	106	53.0%
Relational	9	4.5%
Resource	19	9.5%
Semantic	0	0.0%
Total	200	100.0%

Behavioral propositions (53.0%) dominate, reflecting the emphasis on pre/postcondition specifications in the benchmark. These are the most discriminative for search: two functions with the same type signature but different postconditions (e.g., sorting vs. reversing) are distinguished by their behavioral propositions. Structural propositions (33.0%) capture type-level properties and are used primarily in the type-filtering stage.

11.4 Morphism Type Distribution

Table 8: Morphism Type Distribution in the Semantic Index

Morphism Type	Count	Percentage
Inclusion	12	8.2%
Restriction	91	62.3%
Transport	43	29.5%
Total	146	100.0%

Restriction morphisms dominate (62.3%), reflecting the prevalent pattern of specializing generic containers. Transport morphisms (29.5%) enable cross-program search, linking analogous components in different implementations.

11.5 Domain Breakdown

- **Data Structures:** 5 verified programs, avg. 15.2 props—rich structural constraints enable precise search.
- **Sorting:** 5 verified, avg. 4.8 props—ordering invariants are highly discriminative.
- **Mathematics:** 5 verified, avg. 9.4 props—arithmetic propositions yield high stalk-match scores.
- **String Processing:** 5 verified, avg. 6.4 props—character-level properties provide moderate discrimination.
- **Web/Networking:** 5 verified, avg. 11.2 props—I/O and concurrency constraints are unique to this domain.
- **State Machines:** 5 verified, avg. 15.2 props—transition invariants provide strong discriminative power.

Table 9: Per-Domain Timing (Supplementary Experiments)

Domain	Avg Time	Min Time	Max Time
Sorting	3.59 s	3.18 s	4.00 s
Data Structures	3.41 s	3.27 s	3.75 s
Mathematics	3.76 s	3.15 s	4.05 s
Strings	3.27 s	3.05 s	3.47 s
Web	3.33 s	3.25 s	3.47 s
State Machines	3.81 s	3.41 s	4.30 s

11.6 Equivalence and Refinement Checking

SHEAFSEARCH leverages equivalence checking to identify semantically identical results across different implementations.

Table 10: Equivalence and Refinement Analysis

Metric	Comprehensive	Supplementary
Equivalence Pairs	8	3
Both Verified	8	3
Same Structure	8	—
Mean Equiv. Time	11.06 s	—
Forward Refinement	—	5
Incompatible Pairs	—	5
Total Pairs	—	13

Among supplementary experiments, 5 pairs show forward refinement (one implementation refines the other), 3 pairs are fully equivalent, and 5 pairs are incompatible. The refinement classifica-

tion enables SHEAFSEARCH to deduplicate results: when two candidates are equivalent, we retain only the one with higher trust.

11.7 Scaling Behavior

Table 11 reports timing across different program sizes to characterize the scaling behavior of the search pipeline.

Table 11: End-to-End Timing by Program Size

Size	Mean Time	Min Time	Max Time	Mean Coords
Tiny	4.95 s	4.45 s	5.51 s	3
Small	4.85 s	4.30 s	5.57 s	3.6
Medium	4.47 s	4.26 s	4.74 s	8.6
Large	4.31 s	4.27 s	4.38 s	15

11.8 Bridge Discovery

Cross-program search relies on bridge morphisms discovered during index construction. In our supplementary experiments, 3 bridge packs were discovered, spanning 15 coordinates with 12 inter-program morphisms. These bridges enable queries like “find all implementations of a sorting algorithm” to return results from multiple programs.

11.9 Bug Detection via Search

An unexpected application of sheaf-indexed search is *bug detection*: by searching for programs that *should* satisfy a specification but have obstructions, we identify potential bugs. In our benchmark, 10 programs with known bugs were tested; 10 were correctly identified as matching their buggy specification (the search found them), with mean proposition ratio 1.00 and mean time 5.12 s.

11.10 Analysis

The 100.0% solver-discharge rate across 284 total judgments ensures most results carry formal guarantees, and zero obstructions (0) mean no spurious results. Compared to syntactic search (grep), SHEAFSEARCH avoids false positives by requiring judgment-sheaf entailment. Compared to neural search, SHEAFSEARCH provides formal guarantees at comparable latency.

The cache speedup of 35.0x (from 0.663 ms cold to 0.019 ms warm) makes interactive search practical: after the initial index build, subsequent queries complete in sub-millisecond time. The total experiment time across all 18 programs was 84.3 s (mean 4.68 s, median 4.50 s, stdev 0.45 s).

12 Related Work

Type-Directed Search. Hoogle [1] and Idris search by type signature; we extend this to behavioral specifications and trust levels. Djinn [11] generates code from type signatures using the Curry–Howard correspondence; our approach instead *finds* existing code matching a specification.

Semantic Code Search. CodeBERT [2] and UniXcoder [10] embed code into vector spaces but sacrifice formal guarantees. SHEAFSEARCH provides exact matching on verified properties with approximate fallback via μ . GraphCodeBERT [12] uses data flow graphs as structural features, which is closer to our morphism-based structure but still lacks formal verification.

Specification-Based Retrieval. Pentagraph [3] retrieves by example input-output pairs; we generalize to arbitrary logical specifications. S6 [13] searches Java code by semantic specifications using test-based matching; our approach uses formally verified propositions rather than tests.

Sheaf-Theoretic IR. Robinson [4] and Curry [5] apply sheaves to sensor fusion and topological data analysis; we adapt these ideas to code search over semantic sites. Goguen’s work on sheaf semantics [14] provides the theoretical foundation for using sheaves in software engineering.

Formal Methods in Search. The ISSTA community has explored specification-based testing and retrieval [15]; our approach differs in operating on *verified* specifications rather than test-generated ones. Proof-carrying code [16] attaches proofs to code for safety; SHEAFSEARCH uses analogous certificates for trust-weighted retrieval.

Category Theory in Software Engineering. Fiadeiro and Maibaum [17] use categorical methods for software specification; Goguen [18] develops institutions as a framework for formal specifications. Our work builds on these foundations but applies categorical tools specifically to the retrieval problem.

Topological Data Analysis. Persistent homology and mapper algorithms [19] extract topological features from data; our cohomological obstruction analysis (Section 8) applies similar ideas to diagnose search failures in the semantic index.

13 Conclusion

SHEAFSEARCH demonstrates that semantic sites provide a powerful index for code retrieval. Evaluation on 18 programs shows sub-millisecond latency, 100% pipeline success, and 100.0% solver-level trust.

The categorical foundations (Section 3) reveal that search is naturally a problem in presheaf categories: queries are presheaves, results are sections, and retrieval is mediated by functorial pullback. The Yoneda lemma provides a canonical embedding of coordinates into the search space, and sheafification corresponds to semantic closure.

The descent-theoretic framework (Section 6) enables compositional search: complex queries decompose along coverings, local matches are found independently, and the descent engine glues them into globally consistent results. When gluing fails, the cohomological obstruction analysis (Section 8) provides actionable diagnostics.

Limitations. The current implementation assumes a static corpus; incremental index updates for live codebases remain future work. The covering compatibility score is coarse-grained; finer structural features (e.g., control flow graph similarity) could improve discrimination. The benchmark, while covering 6 domains, is modest in scale compared to industrial code repositories.

Future Work. We identify several directions:

1. **Incremental indexing:** Update the semantic index when source files change, without full reconstruction.
2. **LLM-based query compilation:** Use language models to translate natural language queries into query sheaves.
3. **Repository-scale search:** Scale to millions of coordinates using distributed index sharding.
4. **Higher cohomology:** Use H^2 and higher groups for deeper obstruction analysis.
5. **Cross-language search:** Extend transport morphisms to connect sites across programming languages.

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